

EFA + Reliability Playbook — Your Analysis, Step by Step

Built directly from Dr. Rey's **Assignment #4 guided solution** (the EFA + Reliability procedure on the SAQ dataset), applied to *your* instrument. Follow this in SPSS the moment your data is in. It also doubles as the skeleton of your **Data Analysis** section.

Your parameters (per Dr. Rey's directive):

- **Items analyzed:** the **55 construct items** only (TA_1...RAB_5). *Exclude* the control item (C1), attention checks, demographics, and the open-text from the EFA.
 - **Extraction:** Principal Axis Factoring (PAF)
 - **Rotation:** Direct Oblimin (oblique — your constructs are correlated)
 - **Suppress loadings:** below .30
 - **Target n:** 100 valid
 - **Expected structure:** ~11 factors (11 constructs × 5 items) — but the data decides; report what you find.
 - **Item-removal rule:** remove loadings < .30; for .30–.40 **decide item-by-item** (keep an item if removing it makes the pattern matrix *worse*).
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STEP 0 — Prepare the data (do this first, in SPSS)

- 1 **Get SPSS** — free for FIU students via the FIU software portal (or use SPSS in a campus lab).
 - 2 **Export** your Qualtrics data → SPSS (.sav) or CSV → open in SPSS.
 - 3 **Clean the sample:**
 - Remove anyone who **failed the eligibility screeners** (shouldn't be there, but check).
 - Remove **attention-check failures** (AC1 ≠ "Disagree", AC2 ≠ "Agree").
 - Remove **speeders** (implausibly fast page times) and straight-liners.
 - Document how many you removed → this is your **analytic n**.
 - 1 **■ Reverse-code the (R) items** — item 4 of every construct: TA_4, RA_4, AT_4, SAP_4, FR_4, IR_4, RPG_4, PMI_4, AJQ_4, APR_4, RAB_4.
 - In SPSS: Transform → Recode into Same/Different Variables → 1→5, 2→4, 3→3, 4→2, 5→1 (i.e., new = 6 – old).
 - **This is mandatory before both EFA interpretation and reliability** — Dr. Rey's assignment flags exactly this (their item 3 was reverse-coded).
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STEP 1 — Run the EFA (mirrors the assignment)

Analyze → Dimension Reduction → Factor

- 1 Move **all 55 construct items** into the **Variables** box.

- 2 **Descriptives** button → select **all options** (gives you KMO, Bartlett's, anti-image matrix, correlation matrix, determinant, reproduced correlations).
- 3 **Extraction** button → Method = **Principal axis factoring**; check **Scree plot**.
- 4 **Rotation** button → **Direct Oblimin**.
- 5 **Options** button → **Sorted by size** + **Suppress small coefficients** = **.30**.
- 6 **OK** to run.

STEP 2 — Read the output (in this order)

Output	What to check	Rule of thumb
Descriptives	Means, SDs per item; watch for ceiling effects (very high means, low variance)	Report; flag low-variance items
KMO	Sampling adequacy	≥ .50 minimum, ≥ .70 good, ≥ .80 great
Bartlett's test	Correlations differ from identity matrix	Want significant ($p < .001$)
Anti-image matrix (diagonal)	Per-item sampling adequacy	Each > .50 ; if below, consider removing that item (then rerun)
Correlation matrix + determinant	Multicollinearity	No items uncorrelated with all others (most $r < .3$); none too high ($r > .9$); determinant > .00001
Total Variance Explained + Scree plot	How many factors	eigenvalues > 1, scree inflexion, ≥ 50% variance, and your theoretical expectation (~11)
Reproduced correlations (residuals)	Model fit	Want residuals < .05; concern if > 50% exceed .05
Pattern Matrix	THE table you report	Which items load on which factor (loadings ≥ .30 shown)

STEP 3 — Refine the factor solution

- 1 Look at the **Pattern Matrix**. For each item:
 - **Loads clean (≥ .40 on one factor):** keep.
 - **Loads < .30 everywhere / doesn't show:** remove, then rerun.
 - **Loads .30–.40, or cross-loads on two factors: decide one-by-one** — try removing it; if the pattern matrix gets *cleaner*, drop it; **if it gets worse, keep it** (this is Dr. Rey's exact rule).
- 1 **Rerun after each removal** (removing an item changes everything — KMO, loadings, factors).
- 2 Repeat until you have a clean, interpretable structure.
- 3 **Name each factor** from the items that cluster on it (they should map to your constructs — TA, RA, AT, etc.). Where they don't map cleanly, **that is a validation finding** — report it honestly.

STEP 4 — Reliability (Cronbach's α per factor)

Analyze → Scale → Reliability Analysis

- 1 For **each retained factor**, move its items into the **Items** box.
 - 2 Give the scale a meaningful name (e.g., "Training & Awareness").
 - 3 **Statistics** button → check **Scale if item deleted**.
 - 4 **OK**. Record **Cronbach's α** .
 - $\alpha \geq .70$ = acceptable. If lower, look at "Cronbach's Alpha if Item Deleted" — dropping a weak item may raise it.
 - (Make sure reverse-coded items were recoded in Step 0, or α will be wrong.)
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STEP 5 — Write it up (template from Dr. Rey's model paragraph)

Fill the blanks from your output. Mirror his example almost exactly:

A principal axis factor analysis was conducted on the 55 items with oblique rotation (Direct Oblimin) on a sample of N = ____ valid responses. The Kaiser–Meyer–Olkin measure verified sampling adequacy, KMO = ____ (____ per Kaiser & Rice, 1974), and all individual-item KMO values were greater than ____, above the acceptable limit of .50. Bartlett's test of sphericity, χ^2 (____) = ____, $p < .001$, indicated correlations sufficient for FA. ____ factors had eigenvalues over Kaiser's criterion of 1 and together explained ____% of the variance. The scree plot ____ [supported/was ambiguous about] retaining ____ factors; ____ factors were retained based on _____. The pattern matrix (Table ____) shows the rotated factor loadings. The items clustering on each factor correspond to [Training & Awareness, Rotation of Auditors, ...]. Reliabilities were: [Factor 1] α = ____, [Factor 2] α = ____, ... Items ____ were removed for low or cross-loadings.

Report: the pattern matrix table (always), KMO + Bartlett's, variance explained, the scree plot, and the reliability (α) for each factor. That IS your Data Analysis section — nothing more is required this term (no mediation/hypothesis testing).

What's next for YOU (right now)

- 1 **You can't run this until data is in** — so the critical path is still **data collection to n = 100** (the Sarmed sprint).
- 2 **This playbook is ready and waiting** — the analysis is turnkey the moment you hit 100.
- 3 **Get SPSS installed** now (FIU portal) so it's not a last-minute scramble.
- 4 **Short thank-you to Dr. Rey** for the assignment (draft below).

Dr. Rey — thank you, the guided EFA assignment is exactly what I needed; I've mapped the procedure onto my instrument and will follow it (PAF + Direct Oblimin, suppress $< .30$, your item-removal rule) once collection reaches n = 100. — Yasir

(Reference: procedure adapted from Assignment #4 guided solution; classic EFA reporting per Field, and Kaiser & Rice, 1974, as cited in the assignment.)